

Total marks (41)

Attempt questions 1 – 4

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)

Marks

a) Use the standard integrals to show that:

$$\int_6^{15} \frac{dx}{\sqrt{x^2 + 64}} = \ln 2 \quad 2$$

b) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 2x dx$ (Leave your answer in exact form.) 3

c) Find $\int_1^4 2x\sqrt{1+3x^2} dx$ (Use the substitution $u = 1+3x^2$) 3

d) Show that $\frac{d}{dx}(x^3 e^{x^3}) = 3x^2 e^{x^3} + 3x^5 e^{x^3}$ and use this result to evaluate 3

$$\int_0^1 x^5 e^{x^3} dx.$$

e) Let $f(x) = x + \log_e x$

- i. Show that the curve $y = f(x)$ cuts the x axis between $x = 0.5$ and $x = 1$. 2
- ii. Use one application of Newtons Method with a first approximation to find a second approximation to the root of $x + \log_e x = 0$, to 3 decimal places. 2

Question 2 (8 marks) Begin a new page. Marks

a) Prove that $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{v^2}{2}\right)$. 2

b) A particle is moving in a straight line with $v^2 = 36 - 4x^2$.

i. Prove the particle is moving in simple harmonic motion. 2

ii. What is the amplitude and period of the motion? 2

iii. If the particle is initially at the origin write an expression for its displacement in terms of t . 2

Question 3 (10 marks) Begin a new page.

a) A particle is moving about the origin. Its displacement x centimetres from 0 at time t seconds is given by $x = 3 \cos\left(2t + \frac{\pi}{3}\right)$.

i. Show that the particle is moving under simple harmonic motion. 2

ii. State the amplitude and period of the motion. 2

iii. Find the maximum speed of the motion and when it first occurs. 2

iv. Find the displacement of the particle when it is first at its maximum speed. 2

v. Sketch the displacement-time graph, $0 \leq t \leq \pi$. 2

Question 4 (8 marks) Begin a new page.

a) Fred Kicks a football from ground level at an angle of 72° to the horizontal with velocity 30ms^{-1} . With the origin set where the ball was kicked and using $g = 9.8\text{ms}^{-2}$;

- | | | |
|------|---|---|
| i. | Show the equations of motion are , | 2 |
| | $x = 30t \cos 72^\circ$ | |
| | $y = 30t \sin 72^\circ - 4.9t^2$ | |
| ii. | Calculate the time that the ball is in the air before returning to the ground, to 2 decimal places. | 2 |
| iii. | What is the maximum height the ball reaches, to 3 decimal places? | 1 |
| iv. | If the ball was kicked at an angle of 60° instead of 72° , what change would this make to the distance the ball travelled horizontally before returning to the ground, to 3 decimal places. | 3 |

Extension 1 Assessment 1 Solutions.**Question 1**

a)

$$\int_6^{15} \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

$$\int_6^{15} \frac{dx}{\sqrt{x^2 + 64}} = \left[\ln \left(x + \sqrt{x^2 + 64} \right) \right]_6^{15}$$

$$= \left[\ln \left(15 + \sqrt{15^2 + 64} \right) - \ln \left(6 + \sqrt{6^2 + 64} \right) \right]$$

$$= \ln 32 - \ln 16$$

$$= \ln \frac{32}{16}$$

$$= \ln 2$$

$$\text{b) } \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 2x dx$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{2} \cos 4x + \frac{1}{2} dx$$

$$= \left[\frac{1}{8} \sin 4x + \frac{1}{2} x \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \left[\left(\frac{1}{8} \sin 2\pi + \frac{\pi}{4} \right) - \left(\frac{1}{8} \sin \frac{4\pi}{3} + \frac{\pi}{6} \right) \right]$$

$$= \left[\frac{\pi}{4} - \left(\frac{1}{8} \times \left(-\sin \frac{\pi}{3} \right) + \frac{\pi}{6} \right) \right]$$

$$= \left[\frac{\pi}{4} - \left(-\frac{\sqrt{3}}{16} + \frac{\pi}{6} \right) \right]$$

$$= \frac{12\pi - 8\pi + 3\sqrt{3}}{48}$$

$$= \frac{4\pi - 3\sqrt{3}}{48}$$

$$\text{c) } \int_1^4 2x\sqrt{1+3x^2} dx$$

$$u = 1 + 3x^2$$

$$= \int_4^{49} \sqrt{u} \cdot \frac{1}{3} du$$

$$du = 6x dx$$

$$= \frac{1}{3} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_4^{49}$$

$$x=1, u=4$$

$$= \frac{2}{9} \left[49^{\frac{3}{2}} - 4^{\frac{3}{2}} \right]$$

$$= \frac{2}{9} [343 - 8]$$

$$= \frac{670}{9}$$

$$\begin{aligned} \text{d) } \frac{d(x^2 e^{x^2})}{dx} &= 2x \cdot e^{x^2} + x^2 \cdot 2x e^{x^2} \\ &= 2x e^{x^2} + x^3 e^{x^2} \end{aligned}$$

$$\int_0^1 x^3 e^{x^2} dx = \frac{1}{2} \int_0^1 \frac{d}{dx} (x^2 e^{x^2}) - 2x e^{x^2} dx$$

$$= \frac{1}{2} \left[\left[x^2 e^{x^2} \right]_0^1 - \int_0^1 2x e^{x^2} dx \right]$$

$$= \frac{1}{2} \left[[e - 0]_0^1 - [e^{x^2}]_0^1 \right]$$

$$= \frac{1}{2} [e - [e - 1]]$$

$$= \frac{1}{2}$$

e). i. $f(x) = x + \log_e x$

$$f(0.5) = 0.5 + \log_e 0.5$$

$$= -0.19314718 < 0$$

$$f(1) = 1 + \log_e 1$$

$$= 1 > 0$$

Therefore there is a root between $x = 0.5$ and $x = 1$.

ii.
$$x = 0.5 - \frac{f(0.5)}{f'(0.5)}$$

$$x = 0.5 - \frac{-0.19314718}{1+2}$$

$$x = 0.564382393$$

$$x = 0.564(3dp)$$

Therefore the second approximation for the root is 0.564

Question 2

a) Prove $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$

$$\frac{d^2x}{dt^2} = \frac{dv}{dt}$$

$$\frac{d^2x}{dt^2} = \frac{dv}{dx} \bullet \frac{dx}{dt}$$

$$\frac{d^2x}{dt^2} = \frac{dv}{dx} \bullet V \quad \text{now} \quad V = \frac{d}{dv} \left(\frac{1}{2} v^2 \right)$$

$$\frac{d^2x}{dt^2} = \frac{dv}{dx} \bullet \frac{d}{dv} \left(\frac{1}{2} v^2 \right)$$

$$\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

b) i. $v^2 = 36 - 4x^2$

$$\frac{1}{2}v^2 = 18 - 2x^2$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{d}{dx}(18 - 2x^2)$$

$$\ddot{x} = -4x$$

$$\therefore \ddot{x} = -n^2 x \text{ where } n=2$$

Therefore it moves in SHM

ii $v^2 = 36 - 4x^2$

$$v^2 = -4(x^2 - 9)$$

$$v^2 = -n^2(x^2 - a^2)$$

$$\therefore n = 2 \text{ and } a = \pm 3$$

Therefore the amplitude is 3 and the period is $-\frac{2\pi}{2} = \pi$

iii $t = 0$ and $x = 0$

$$v^2 = 36 - 4x^2$$

OR consider this solution

$$v = \sqrt{36 - 4x^2}$$

From ii. $N=2$ and $a=3$

$$\frac{dx}{dt} = \sqrt{36 - 4x^2}$$

a possible solution is $x = 3 \sin 2t$

$$\frac{dt}{dx} = \frac{1}{\sqrt{36 - 4x^2}}$$

CHECK

$$\dot{x} = 6 \cos 2t$$

$$\int dt = \int \frac{1}{2\sqrt{9 - x^2}} dx$$

$$\ddot{x} = -12 \sin 2t$$

$$\therefore t = \frac{1}{2} \sin^{-1} \frac{x}{3} + C$$

$$\ddot{x} = -4(3 \sin 2t)$$

As $t = 0$ and $x = 0$ then $C = 0$

$$\ddot{x} = -4x$$

$$\therefore 2t = \sin^{-1} \frac{x}{3}$$

which satisfies the differential eqn.

$\therefore x = 3 \sin 2t$ is the solution

$$\sin 2t = \frac{x}{3}$$

$$x = 3 \sin 2t$$

Question 3

a) i. $x = 3 \cos\left(2t + \frac{\pi}{3}\right)$

$$\dot{x} = -6 \sin\left(2t + \frac{\pi}{3}\right)$$

$$\ddot{x} = -12 \cos\left(2t + \frac{\pi}{3}\right)$$

$$\ddot{x} = -4\left(3 \cos\left(2t + \frac{\pi}{3}\right)\right)$$

$$\ddot{x} = -4x \quad \text{where } x = 3 \cos\left(2t + \frac{\pi}{3}\right)$$

$$\ddot{x} = -n^2 x$$

$$\text{where } n=2$$

Therefore it is moving in SHM.

li Amplitude = 3

$$\frac{2\pi}{\text{Period}} = \pi$$

iii. max speed = velocity

Max value of $\sin\left(2t + \frac{\pi}{3}\right)$ is ± 1

Max speed = $|-6 \times 1| = 6 \text{ m/s}$

$$6 = \left| -6 \sin\left(2t + \frac{\pi}{3}\right) \right|$$

$$1 = \sin\left(2t + \frac{\pi}{3}\right)$$

$$\therefore \left(2t + \frac{\pi}{3}\right) = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$$

$$t = \frac{\pi}{12}, \frac{13\pi}{12}, \dots$$

Therefore the first time it is at max speed is $\frac{\pi}{12}$ s.

iv. $x = 3 \cos\left(2t + \frac{\pi}{3}\right),$

$$x = 3 \cos\left(2 \cdot \frac{\pi}{12} + \frac{\pi}{3}\right)$$

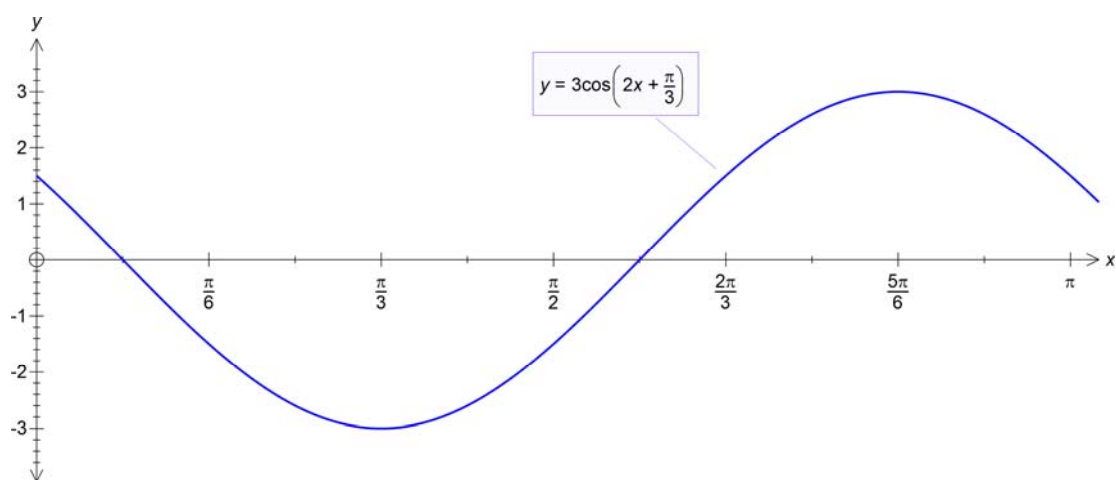
$$x = 3 \cos\left(\frac{\pi}{2}\right)$$

$$x = 0 \text{ m.}$$

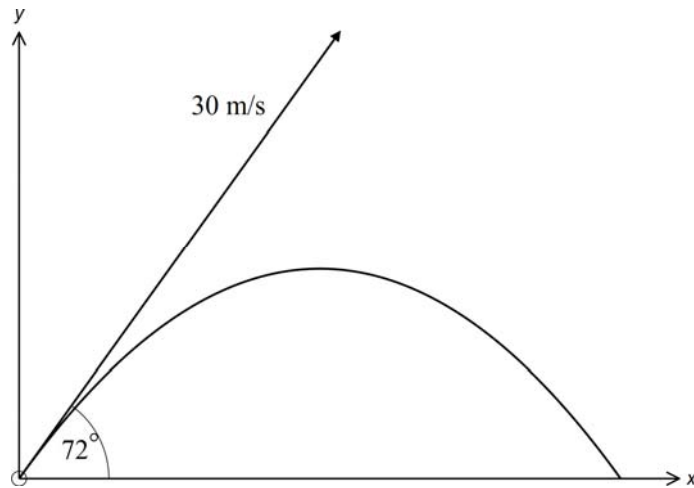
v.

Sketch

$$x = 3 \cos\left(2t + \frac{\pi}{3}\right)$$



Question 4



Initial velocity

$$\cos 72^\circ = \frac{\dot{x}}{30}$$

$$\dot{x} = 30 \cos 72^\circ$$

$$\sin 72^\circ = \frac{\dot{y}}{30}$$

$$\dot{y} = 30 \sin 72^\circ$$

Horizontal Motion

$$\ddot{x} = 0$$

$$\dot{x} = C_1 \quad t = 0, \quad \dot{x} = 30 \cos 72^\circ \therefore C_1 = 30 \cos 72^\circ$$

$$\therefore \dot{x} = 30 \cos 72^\circ$$

$$\therefore x = 30 \cos 72^\circ t + C_2 \quad t = 0, x = 0, \therefore C_2 = 0$$

$$\therefore x = 30 \cos 72^\circ t$$

Vertical Motion

$$\ddot{y} = -9.8$$

$$\dot{y} = -9.8t + C_3 \quad t = 0, \dot{y} = 30 \sin 72^\circ, C_3 = 30 \sin 72^\circ$$

$$\dot{y} = -9.8t + 30 \sin 72^\circ$$

$$y = -4.9t^2 + 30 \sin 72^\circ t + C_4 \quad t = 0, y = 0, \therefore C_4 = 0$$

$$\therefore y = -4.9t^2 + 30 \sin 72^\circ t$$

b) $y = 0, t = ?$

$$\therefore -4.9t^2 + 30 \sin 72^\circ t = 0$$

$$\therefore t(-4.9t + 30 \sin 72^\circ) = 0$$

$$\therefore t = 0 \quad \text{or} \quad \therefore -4.9t + 30 \sin 72^\circ = 0$$

$$\therefore t = \frac{30}{4.9} \sin 72^\circ$$

$$t = 5.82279499$$

$$t = 5.82s$$

Therefore the ball returns to the ground after 5.82 seconds.

c) Max height

Max height occurs at $t = 5.82279499 \div 2s$

$$t = 2.91139749s$$

$$\therefore y = -4.9(2.91139749)^2 + 30 \sin 72^\circ (2.91139749)$$

$$y = 41.5335344$$

$$y = 42m(n.m)$$

d) horizontal distance difference.

$$\therefore x = 30 \cos 72^\circ t \quad t = 5.82279499$$

$$\therefore x = 30 \cos 60^\circ \times 5.82279499$$

$$x = 87.3419248$$

$$\therefore x = 30 \cos 60^\circ t$$

$$\therefore x = 30 \cos 72^\circ \times 5.82279499$$

$$x = 50.04247004$$

$$\text{Difference} = 37.29945481$$

$$= 37m(n.m)$$

Therefore the difference in length is 37m